

COP 3515 – Advanced Program Design

Semester Programming Project

Client Change Request (CCR-001)

Project: Student Information Management System (SIMS)

Customer: University Registrar's Office

Priority: High

Assigned To: Student Developer

Sprint: Week 1

Due Date: Beginning of Lesson 4

Business Background

The University Registrar's Office currently maintains student information using several spreadsheets and paper forms. This process is time-consuming and often leads to duplicated information and data entry errors.

The university has contracted your software development team to build a command-line application named the **Student Information Management System (SIMS)**. Because of budget limitations, the system will be developed over multiple weeks. Each week the Registrar will submit a new Client Change Request (CCR) describing additional functionality they would like added to the system.

For this first sprint, the Registrar would like a very simple prototype that demonstrates basic data entry and formatted output.

Business Requirements

The Registrar would like the program to:

1. Display the course title.
2. Display the programmer's name.
3. Display the current version number of the software.
4. Display a welcome message for the user.

5. Prompt the user to enter:

- Student ID
- Student Name
- Current GPA

6. Display a neatly formatted summary of the information entered.

At this stage, the program only needs to process information for **one student**.

Acceptance Criteria

The project will be considered complete when all of the following requirements have been met.

- ☐ The program compiles without errors or warnings.
 - ☐ The program displays the course title.
 - ☐ The program displays the programmer's name.
 - ☐ The program displays a version number.
 - ☐ The program displays a welcome message.
 - ☐ The user can enter a student ID.
 - ☐ The user can enter a student name.
 - ☐ The user can enter a GPA.
 - ☐ The program displays all entered information in a clear and readable format.
 - ☐ Output is properly aligned and professionally formatted.
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Technical Constraints

For this week's assignment, you may only use the C language features covered in class.

Permitted concepts include:

- `printf()`
- Variables
- Constants
- Boolean variables
- Basic input/output
- Formatted output

Do **not** use language features that have not yet been introduced in class.

Out of Scope

The Registrar is **not** requesting any of the following features at this time.

- Saving data to a file
- Multiple student records
- Arrays
- Functions
- Loops
- Menus
- Searching
- Editing records
- Dynamic memory allocation

These features will be requested in future Client Change Requests.

Example Program Execution

Example 1

User Input

Student ID: 1234567

Student Name: Alice Johnson

Current GPA: 3.84

Expected Output

Student Information Management System

Version 1.0

Programmer: Jane Smith

Welcome to SIMS

Student Summary

Student ID : 1234567

Student Name : Alice Johnson

Current GPA : 3.84

Example 2

User Input

Student ID: 1002003

Student Name: Michael Brown

Current GPA: 2.91

Expected Output

Student Summary

Student ID : 1002003

Student Name : Michael Brown

Current GPA : 2.91

Testing Requirements

Before submitting your assignment, verify that your program works correctly.

Your submission must include evidence that you tested at least the following three scenarios.

Test	Student ID	Student Name	GPA
Test 1	1234567	Alice Johnson	3.84
Test 2	1002003	Michael Brown	2.91
Test 3	9876543	Christopher Williams	4.00

For each test, include:

- Input entered
 - Program output
 - Pass/Fail result
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AI Usage Policy

Artificial Intelligence tools (such as ChatGPT, Copilot, Claude, or Gemini) **may be used** during this assignment.

However, **you are responsible for all submitted work.**

If AI-generated code contains errors, those errors are your responsibility.

Your grade is based on the correctness of your submitted program, not on how the code was produced.

Submission Requirements

Submit the following files to Canvas in a single ZIP file:

- main.c
- Source code for any additional files (if applicable)
- A PDF containing:
 - Screenshots of your three required test cases
 - A brief description of any bugs you encountered and how you corrected them

Looking Ahead

This is the first release of the **Student Information Management System (SIMS)**.

Future Client Change Requests will require you to **modify this existing codebase** rather than starting over. As the semester progresses, you will add features such as grade management, file storage, functions, pointers, dynamic memory allocation, and command-line processing.

Important: Keep your code well organized. Future assignments will build directly on the work you complete this week.